



Tackling common issues in hydraulic systems

How the right lubricant and
maintenance decisions can help
you prevent breakdowns



SHELL
LUBRICANT
SOLUTIONS

Foreword

Your hydraulic system provides the power and precision that the majority of metals processes command. However, only high-quality lubrication can enable these systems to operate effectively and consistently amid the extreme pressures and temperatures that metals production involves.

Today, the operating environment that your hydraulic fluid must perform within is more demanding than ever – largely due to:

- More compact and connected equipment
- Increasing operating pressures and temperatures
- Greater power and efficiency requirements
- More and stricter environmental regulations

These immediate and emerging market trends make your choice of oil an increasingly important one if you want to avoid costly maintenance and unplanned delays. Therefore, it's perhaps time to delve a little deeper into the key issues that are often responsible for hydraulics system breakdowns.¹

Breakdowns that, more often than not, relate to issues such as sludge and varnish formation, water contamination and aeration. All of which can be overcome by the correct lubricant and effective maintenance practices, as explored over the next few pages.

Cem Ayaz,
Global Metal Sector Manager,
Shell Lubricant Solutions



Three common hydraulic system challenges facing the metals industry:



01

Sludge and varnish

How to deal with the thermal degradation and oxidation of hydraulic fluids

02

Water contamination

Avoiding rust, leakage and water's effect on fatigue strength

03

Aeration

Preventing component damage, reduced efficiency, and increased susceptibility to harmful system malfunctions

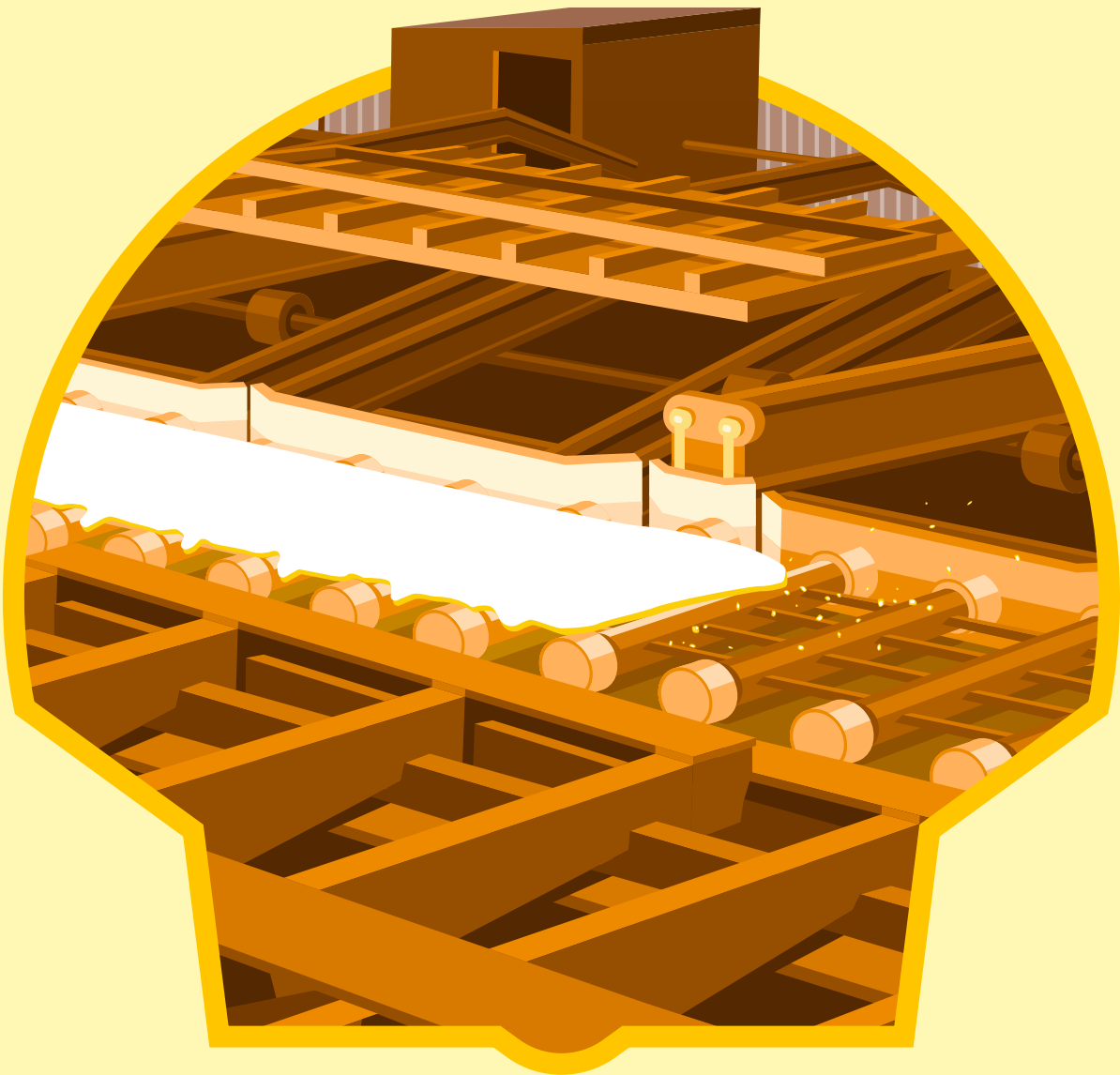
01

Sludge and varnish

When lubricant degrades, two products are left within the system: sludge and varnish. Both examples of deposits, sludge is a gummy, soft and mud-like substance that settles out of the oil and can sometimes lead to varnish – a thin, orange, brown or black film that occurs on the interior of lubricant systems.

Varnish is difficult to remove by manually wiping alone – and often going unnoticed until equipment failure – these deposits require immediate remedial action to prevent them from affecting either fluid flow or mechanical movement. Which, in turn, can contribute to wear and corrosion throughout hydraulic systems.

Issue	Implication
Servo valve sticking	Unit downtime and outage
Heat exchange plate out	Increased oil temperature and oxidation
Increased bearing temperatures	Reduced bearing life, premature bearing failure
Excessive gear wear	Reduced gear life, premature gear failure
Additive depletion	Decreased oil life, higher costs
O-ring and seal failure	Oil leaks and cylinder problems
Suction strainer clogging	Pump failures





Main causes of sludge and varnish

There are two main causes of these types of deposit formations. The first is **oxidation**, which is a chemical breakdown of oil that occurs in the presence of oxygen and can be accelerated by high temperatures, water, acids and catalysts like copper. In fact, the Arrhenius Rate Rule suggests (once a lubricant exceeds its base activation temperature) every 10°C (18°F) that operating temperatures rise, the oxidation rate doubles.² The second reason is **thermal degradation**, a similar process that occurs without oxygen present, and can be caused by spark discharge, micro-dieseling and hot spots.

Dealing with oxidation

To overcome oxidation issues in hydraulic systems, a fluid with strong oxidation resistance can potentially help to enable longer fluid life, while restricting oil thickening and the formation of acid. This usually stems from a fluid's high quality base oil (for example, Group II and above have better antioxidation properties) as well as the additives present in its formulation. Together, this combination can help extend fluid-drain intervals and reduce maintenance requirements across the entire chain. All of which can keep costs under control, especially when you consider the alternative choices of either changing lubricant more frequently, reducing overall equipment use or increasing maintenance scheduling.

Dealing with thermal degradation

Given that high operating temperatures can increase the likelihood of oil degradation, and therefore sludge formation, it is vital for maintenance managers to practise proactive, preventive action before it can affect overall hydraulic system efficiency. This means actively choosing a fluid with good thermal stability that can minimise sludge build-up in the reservoir, which could otherwise lead to equipment wear or shortened filter and oil life.



The role of oil analysis, testing and varnish removal

What makes the remedial process tougher still – and the role of lubricant even more important – is the fact that varnish is not always detectable through traditional oil analysis. This is because these issues are often localised in one specific area, rather than affecting the entire oil. As a result, a system's viscosity, FTIR-oxidation or Total Acid Number (TAN) can appear unchanged, despite there being high varnish potential.

With that being said, there are a few tests that are worth considering, such as the membrane patch colorimetry (MPC)

offered by Shell LubeAnalyst – our oil and equipment monitoring service that provides access to more than 100 different tests. MPC provides a photometric evaluation of the colour intensity of a test membrane following filtration, indicating where possible deposits may exist. Ultimately, this is a quick and cost-effective way of assessing your oil's varnish potential. A varnish removal service, as provided by Shell LubeOptimiser, can then improve system performance by removing insoluble sticky deposits from critical components, enabling the effective flow of oil around the system.

Top tips for dealing with sludge and varnish issues

[See p.7 to discover the benefits of Shell's hydraulic fluids](#)



Practise preventive or, better yet, predictive maintenance



Choose a hydraulic fluid with good thermal stability



Extend fluid-drain intervals by using oxidation-resistant fluids



02

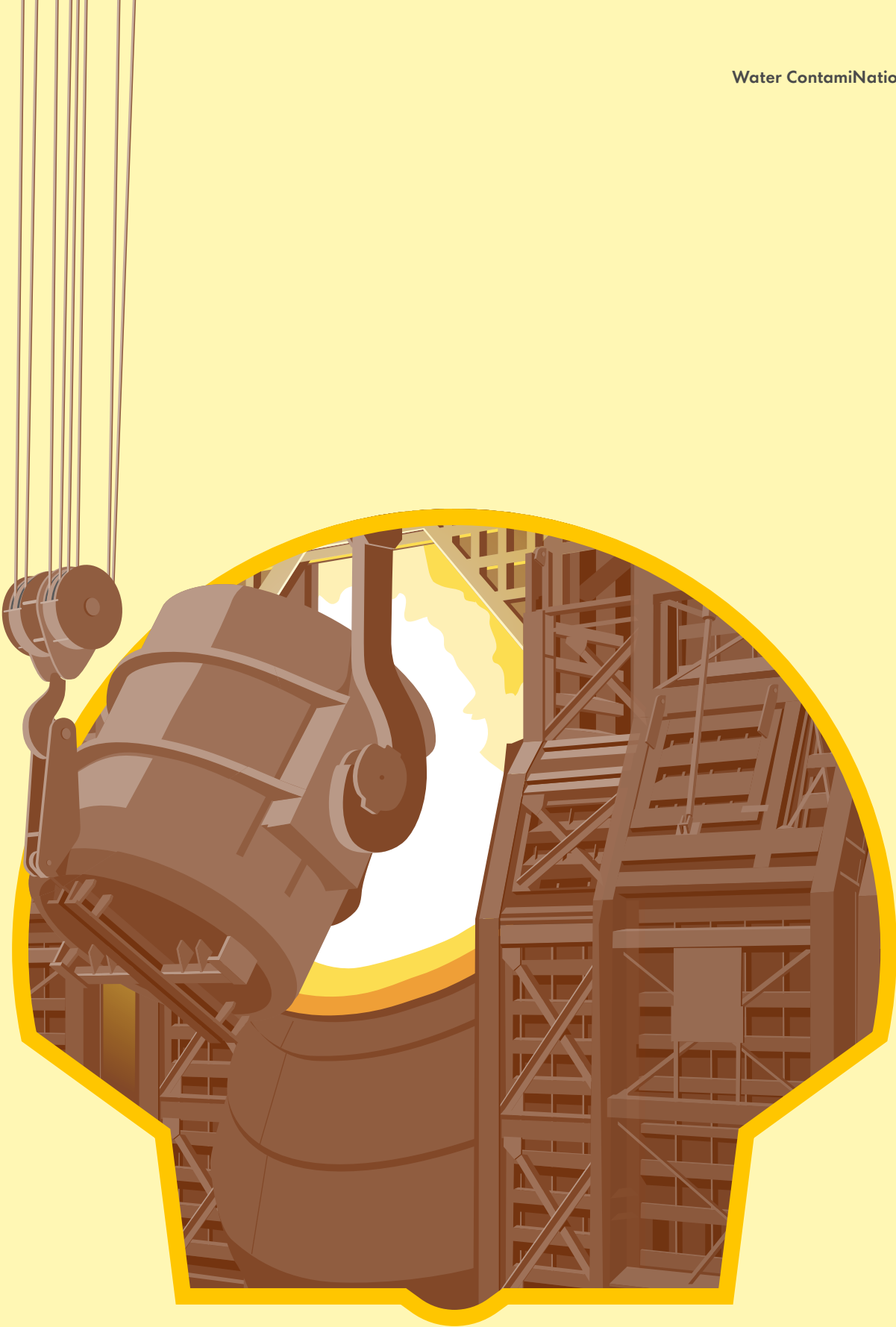
Water Contamination

Alongside heat and oxygen, water is one of the most challenging elements for hydraulic systems to deal with. Worse still, it is not a contaminant that can easily be prevented given the multiple pathways it has for entering a system, between condensation via vents or covers, oil cooler leakage and incorrectly stored fluid.

The potential consequences of moisture contamination are similarly wide-ranging, though the majority of these stem from the onset of rust and subsequent mechanical problems. In addition to the destructive effect of surface wear, rust particles can also lead to a number of detrimental issues.

Issue		Implication	
Formation of oxidation sludge and deposits	⚠️	Serious malfunctioning of critical components	⊗
Rust particles in fluid circulation		Abrasive wear	
Accumulation of rust against dynamic seals		Accelerated wear and increased fluid leakage	
Hydrolysis reactions with additives		Poor demulsibility and potential filter blocking	

But rust isn't the only damaging side effect of water contamination. For hydraulic motors and pumps that rely on heavily stressed ball and roller bearings or gears, water's effect on fatigue strength can be significant; in fact, tests confirm that as little as 0.01% (100ppm) of dissolved water can reduce fatigue life by up to 48%.³ Meanwhile, general water ingress can also cause: component corrosion, deterioration of lubricating film properties, accelerated oil oxidation rate, additive breakdown, bacterial growth, and general lubricant degradation.

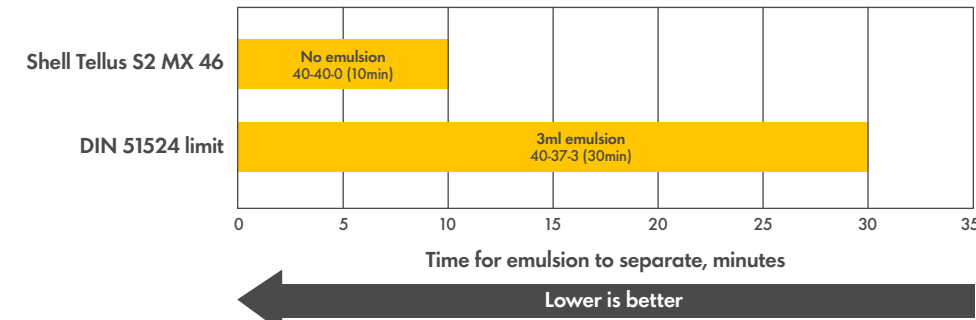


How to deal with water contamination

While there are some component-based remedies for mitigating water contamination, such as installing a drain cock at the bottom of fluid reservoirs or using special filter elements, the most effective solution is almost always a hydraulic oil with good demulsibility and rust protection. However, while the ability to protect components from rust should be a basic property for any hydraulic fluid, many rust inhibitors actually detract from the hydrophobic qualities that

highly refined mineral base oils – or certain synthetic oils – possess. As a result, it's not always easy to find a fluid that can balance rust protection with the capacity to separate rapidly from entrained water during stationary periods at the fluid reservoir. This is especially true when you factor in the contamination potential of soluble emulsifiers or metal soaps – both of which, while widely used, can negatively affect a fluid's demulsibility.

Benefit from water shedding (ASTM D1401) at 54°C



10 minutes is the typical value measured in the lab. 20 minutes is the maximum value that customers can expect.

- ✓ Excellent resistance to breakdown in presence of water
 - ✓ Excellent filterability* even under wet conditions
 - ✓ Consistent water separation** and air release*** ensure efficient operation throughout your hydraulic system.
- * Compared with ISO 13357-1 filterability test limit.
** Compared to water separation ASTM D1401 limit.
*** Compared with IP 313 air release limit

The importance of hydrolytic stability

However, sometimes, a degree of water contamination is unavoidable. If this occurs, then the issue moves from one of demulsibility to one of hydrolytic stability – that is, a lubricant's ability to resist degradation in the presence of water. Good hydrolytic stability ultimately prevents water contamination from limiting fluid life, while it also ensures

fluid retains its desirable properties, like those provided by additives. As a result, it can help extend your fluid-drain interval, reduce maintenance requirements and limit unnecessary costs. With 55% of lubrication professionals having witnessed the effects of hydrolysis in machines at their plants, this is evidently an important lubricant characteristic.⁴

Top tips for dealing with water contamination issues

See p.7 to discover the benefits of Shell's hydraulic fluids

- Choose a hydraulic fluid with good demulsibility and rust protection
- Consider the consequences of soluble emulsifiers or metal soaps
- Prevent fluid life from being limited by deploying a fluid with good hydrolytic stability

03

Aeration

When air begins circulating, or dissolving, in a hydraulic fluid, that fluid’s protective properties can quickly degrade. From high noise levels to unreliable operation, the resulting consequences on the hydraulic systems can be both quick to appear and hard to ignore.

Ultimately, aeration in the oil can lead to undesirable consequences such as component damage, reduced efficiency and responsiveness, as well as increased susceptibility to erosion, cavitation or other harmful system malfunctions. All of which can cause unplanned downtime and increase maintenance costs.

Issue	Implication
Undissolved air	Cavitation on suction side of hydraulic pumps
Pressure drop caused by undissolved air	Laminar flow replaced with turbulent conditions
Collapsing air bubbles on pressure side	Damage resembling that of cavitation, oil dieseling and system noise
Free air bubbles passing through hydraulic pump	Temperature rise, potential thermal degradation
Foaming	Oil level control issues, air locks at high points





Main causes of aeration

There is not one, single cause of aeration in the hydraulic system, which can often make it extremely difficult to diagnose. Worse still, an incorrect diagnosis – and remedy – can further aggravate the issue. Everything from leaks around pump seals and pipe fittings to the dispersion of finely divided air bubbles can contribute to air circulation within the system.

However, modern hydraulic systems pose a further challenge still. Reduced reservoir or sump sizes, high flow rates and increased operating pressures all make today's equipment more susceptible to oil aeration. This may be a result of return fluid, containing free air, splashing freely into the fluid reservoir or it could be due to low fluid level in the reservoir itself. With residence times as short as 30 seconds in some of these lower-volume reservoirs, there can be insufficient time allowed for vital air release.⁵



How to deal with aeration

The best way to tackle aeration is to identify the source of the issue, before making the necessary changes to your equipment to prevent or slow the air ingress that is occurring. But, because aeration exacerbates turbulent system or fluid conditions, the next step is to use a hydraulic fluid with better air release properties. While a lubricant with poor air release properties may exhibit increased foaming, leading to defective valve, hydraulic motor and actuator operation, good air release on the other hand can help ensure clean oil, low pressure loss and efficient operation.



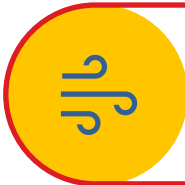
Gas-to-liquids (GTL) technology

Air release properties are predominantly dictated by a fluid's base oil. Shell Tellus S4 VE, for example, has a 27% lower air release time when compared with a competitor's mineral oil.⁶ That is largely due to the GTL technology, with tests recording significantly faster air release when compared with mineral-based oils, resulting in less entrained air and up to 8% greater volumetric efficiency – in addition to a 50% perceived reduction in sound.⁵

[See p.8 to discover the benefits of GTL technology](#)

Top tips for dealing with aeration issues

[See p.7 to discover the benefits of Shell's hydraulic fluids](#)

-  Ensure chosen hydraulic fluid has good air release qualities
-  Consider a low viscosity fluid, which produces larger foam bubbles
-  Explore GTL-based fluid as an alternative to mineral-based fluid

Overview of Shell hydraulics products and solution

Shell Lubricant Solutions has a wide range of hydraulic fluids formulated to meet the needs of today’s hydraulic systems, as well as services that can help unlock added value by reducing downtime.



Shell hydraulic fluid portfolio

Product specification

Product	Benefits	Technology
Shell Tellus S4 VE	Energy efficiency Extended oil life Sludge and varnish control	GTL synthetic zinc-based
Shell Tellus S4 ME	Extra long life Energy saving	Synthetic, ashless
Shell Tellus S4 VK	Ultra-low temperature Versatile application Shear stable	GTL synthetic zinc-based
Shell Tellus S4 VX	Ultra-low temperature Versatile application	Mineral, ashless
Shell Tellus S3 Z	Longer oil life Enhanced protection Industrial applications	Group II/GTL synthetic blend
Shell Tellus S3 V	Long life and improved efficiency Versatile application	Group II mineral,ashless
Shell Tellus S3 M	Long life and improved protection Industrial applications	Group II mineral, zinc free
Shell Tellus S2 VX	Extra protection Versatile application Long life	Group II mineral, zinc-based
Shell Tellus S2 MX	Extra protection Industrial applications Long life	Group II mineral, zinc-based
Shell Hydraulic S1 M	Reliable protection Industrial applications	Mineral, zinc-based



Value-adding services

Our technical services can help to unlock substantial value by reducing downtime. Through these services, we can work with you to ensure you have the right products, applied in the right ways, with the right monitoring and analysis programmes to keep your equipment operating efficiently.

Shell LubeAnalyst	Predictive oil condition monitoring program
Shell LubeCoach	Modular lubricant training program delivered by Shell Experts face-to-face or available online
Shell LubeAdvisor	Site Assessment, Detailed Lubrication survey and Technical support to enable total cost of ownership improvements
Shell LubeOptimiser	Field services including filtration, varnish removal and vacuum dehydration
Shell LubeChat	Brings you a vast array of Shell knowledge, equipping you to make faster, better-informed decisions that can ultimately help bring down your total cost of ownership



Shell Tellus S4 VE and GTL technology

To meet the growing demands of today’s increasingly harsh operating environments, Shell has developed a range of special, high-viscosity-index base oils using GTL technology.

GTL technology works by converting natural gas into high quality liquid products – like synthetic base oils – with almost none of the impurities found in those made from crude oil. With almost half a century of research behind them, Shell’s GTL Base Oils deliver several benefits.

Feature		Benefit	
Special, high viscosity index	⚙️	Minimises oil’s viscosity variation, exhibiting good fluidity amid fluctuating temperatures	📈
Little need for VI improvers		Increases shear stability, especially for ISO VG 32 and ISO VG 46 grades (5% KRL shear max.)	
Low shear rate		Helps maintain pump performance, even in highly loaded equipment	
Exceptional surface properties (e.g. fast aeration)		Higher system efficiency, less noise, less cavitation	
High performance additive technology		Exceptional stick-slip performance	

The Shell Tellus S4 VE range of advanced and energy efficient, synthetic hydraulic fluids is a perfect example of how GTL technology is helping to shape the future of hydraulics, by:

5x

Delivering more than 5 times lower sludge compared with ISO HV Synthetic Competitor Oil A**

10,000 hours

Exceeding the 10,000 hours maximum duration that can be measured in the industry-standard turbine oil stability test (TOST)

27%

Providing a 27% faster air release time when compared with a competitor’s mineral oil***

10x

Offering more than 10 times less sludge than a competitor’s mineral oil (Dry TOST ASTM D 7873)****

6%

Delivering a 6% hydraulic productivity improvement*

21%

Providing up to a 21% reduction in energy loss from hydraulic pumps*

All of which results in: less equipment wear; reduced fuel consumption; better equipment cleanliness; longer lubricant life; and lower maintenance costs.

* When compared to mineral-oil-based fluid.
** Based on VG 46, based on Shell internal competitor testing.
*** GTL-based fluids have significantly faster air release compared with the mineral-based oils, which means less entrained air and up to 8% greater volumetric efficiency.
**** Measured using industry standard, Shell proprietary and internal competitor benchmarking tests. Actual effects and benefits may vary. No guarantees provided. VG 46, based on Shell internal competitor testing.



Shell
Tellus

Shell Tellus in action:
a real-world, global impact

	Application	Challenge	Solutions	Impact	Est. cost saving
A Chinese steel manufacturer	Danieli high-speed wire rod mill hydraulic system	Frequent unplanned downtime caused by: short seal life, serious oil leakage, and sludge formation blocking valves	Shell LubeAdvisor Shell Tellus S2 M 46	- Reduced corrosion, oil leakage and sludge formation - Increased oil, component and seal life - Improved system reliability through no unplanned downtime	\$2,868,600 / year
A Chinese steel manufacturer	Zhengzhou Hengxu rolling equipment	- Insufficient reliability, uptime and productivity - Frequent line stoppages for maintenance checks caused by blocked filters and valves	Shell LubeAnalyst Shell Tellus S3 M 46	- Improved performance - Significant reduction in filter and valve build-up - Extended fluid-drain interval from four to eight months (100% ODI increase)	\$47,000 / year
A German steel manufacturer	Simplex hydraulics	Cylinder seal leakage preventing 100% equipment availability	Shell Tellus S4 ME 46	- Protected cylinder seals - Eliminated downtime caused by leakage and varnish build-up - Able to safely extend ODI 41% reduction in operating costs	\$39,000 / year

Endnotes

¹ McKinsey & Company. [“Reengineering construction equipment: from operations focused to customer centric.”](#) April 2016.

² [Machinery Lubrication: Advice for extending Lubricant Life](#)

³ Peter Hodges. “Hydraulic Fluids.” Elsevier. June 1996.

⁴ Wes Cash. [“Understanding hydrolysis and hydrolytic stability.”](#) Machinery Lubrication. June 2013.

⁵ Shell. [“Hydraulic power efficiency: Evaluating the gas-to-liquids base oil advantage.”](#)

⁶ Measured using industry-standard, third-party and internal competitor benchmarking tests. Actual effects and benefits may vary. No guarantees provided.



For more information on how to improve energy efficiency and enhance equipment protection through lubrication, visit:

[Shell.com/Metals](https://www.shell.com/metals)

